

Ali Meski

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EDUCATION

Bachelor of Engineering, Systems and Computing Engineering (Co-op)

2024–Present

University of Guelph, Guelph, Ontario

- Dean's Honours List – Awarded for achieving an average of 80% in the Winter 2025 semester
- Recipient of Entrance Scholarship of \$2000 for admission average of 93%
- Relevant Courses: Engineering Design 2, Fluid Mechanics, Engineering Systems Analysis

TECHNICAL SKILLS

Embedded & Edge AI: Raspberry Pi 5, Hailo NPU model compilation & quantization (DFC), STM32 (HAL, interrupt-driven UART/I2C), C/C++, sensor integration, VHDL, MATLAB

Robotics & Sensors: ROS2, SLAM (slam_toolbox), LiDAR, stereo depth cameras, ToF sensors, motor control, real-time state machine design

Mechanical & Design: AutoCAD, SolidWorks, mechanical system integration, system control logic

Software: Python, PyTorch, Ultralytics YOLO, OpenCV, Java, Kotlin, SQL, Git, Docker

Tools: VS Code, CLion, IntelliJ, PyCharm, Google Cloud Platform, Microsoft Office

EXPERIENCE

Robotics Research Assistant

May 2025 – Present

University of Guelph, Dept. of Electrical and Computer Engineering

Guelph, Ontario

- Built a real-time **fall detection pipeline**, compiling a pose model (YOLO26n-pose) to Hailo Executable Format and deploying it on a Hailo-10H NPU (40 TOPS) alongside a Raspberry Pi 5 and STM32G474RE.
- Diagnosed a model quantization failure collapsing class confidence output; fixed it via representative calibration data.
- Integrated a stereo depth camera, RPLIDAR, and dual ToF sensors into a unified sensing stack for obstacle-aware navigation.
- Built a ROS2 publisher feeding live person-tracking data into a collaborator's LiDAR-based path planning system.
- Wrote embedded C firmware on an STM32G474RE (interrupt-driven UART/I2C, HAL) for microcontroller-level sensor handling.
- Co-authored an **IEEE-format conference paper** on the system architecture, restructured the team's GitHub codebase, and maintained weekly progress logs.

PROJECTS

Autonomous Slingshot Rover | C++, AutoCAD, Arduino

- Designed and built an Arduino-based autonomous rover integrating mechanical, electronic, and software subsystems for real-world deployment and targeting.
- Integrated ultrasonic and proximity sensors with servo actuation to enable obstacle detection, autonomous navigation, and automated slingshot firing.
- Developed control logic and system architecture from schematic to physical build, ensuring reliable hardware-software integration under operational conditions.

Hydraulic Robot CAD Modelling & Mechanical Analysis | SolidWorks

- Reverse engineered a multi-configuration hydraulic robotic machine by creating precise 3D SolidWorks CAD models from physical measurements, capturing real-world geometry across 18+ modelled features.
- Produced detailed 2D engineering drawings (ISO A4) with correct tolerancing and assembled all components into a full 3D CAD animation demonstrating subsystem motion including gears, pistons, and linkages.

Game Boy Emulator | C, SDL2

- Engineered a fully working Game Boy emulator replicating CPU, memory, PPU, APU, and input subsystems with cycle-accurate instruction decoding for full ROM compatibility.
- Architected a modular system with SDL2 for real-time graphics rendering, enhancing maintainability and enabling scalability for additional emulator features.